

Supplementary Materials: Robust Tensor Learning with Graph Diffusion for Scalable Multi-view Graph Clustering

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In this supplementary material, we provide comprehensive proofs for the theorems proposed in the main manuscript. The document is structured as follows:

- Section A offers a detailed elaboration on the proof of Theorem 1.
- Section B provides an in-depth analysis and proof of Theorem 2.
- Section C presents a thorough and rigorous proof of Theorem 3.

1 A: Proofs of Theorem 1

THEOREM 1. Suppose the graph diffusion matrix $\mathbf{F} \in \mathbb{R}^{n \times n}$ is derived from the bipartite graph $\mathbf{B} \in \mathbb{R}^{n \times m}$. The graph diffusion matrix can be expressed as follows:

$$\mathbf{F} = e^{-\eta} \mathbf{I}_n - \hat{\mathbf{P}} \hat{\mathbf{P}}^T (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T, \quad (1)$$

where $\mathbf{P} = \mathbf{B} \mathbf{D}^{-\frac{1}{2}}$, $\mathbf{D}$ is the column degree matrix of $\mathbf{B}$, defined as $\mathbf{D} = \text{diag}(\mathbf{B} \mathbf{1})$, $\hat{\mathbf{P}} = e^{-\eta} \mathbf{I}_m - \exp(-\eta \mathbf{L}_m)$, $\mathbf{L}_m = \mathbf{I}_m - \mathbf{P}^T \mathbf{P}$

PROOF. Consider the graph diffusion matrix $\mathbf{F} \in \mathbb{R}^{n \times n}$ derived from the bipartite graph $\mathbf{B} \in \mathbb{R}^{n \times m}$. Let $\mathbf{P} = \mathbf{B} \mathbf{D}^{-\frac{1}{2}}$, where $\mathbf{D} = \text{diag}(\mathbf{B} \mathbf{1})$ is the column degree matrix of $\mathbf{B}$. As presented in the main manuscript, the graph diffusion matrix can be expressed as:

$$\begin{aligned} \mathbf{F} &= e^{-\eta} \sum_{t=0}^{\infty} \left(\frac{\eta^t}{t!} (\mathbf{P} \mathbf{P}^T)^t \right) \\ &= e^{-\eta} \times \frac{\eta^0}{0!} \times \mathbf{I}_n - e^{-\eta} \times \frac{\eta^0}{0!} \times \mathbf{P} (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \\ &\quad + \mathbf{P} \times \left[e^{-\eta} \sum_{t=0}^{\infty} \left(\frac{\eta^t}{t!} (\mathbf{P}^T \mathbf{P})^t \right) \right] \times (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \end{aligned} \quad (2)$$

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This can be further simplified to:

$$\begin{aligned} \mathbf{F} &= e^{-\eta} \times \frac{\eta^0}{0!} \times \mathbf{I}_n - e^{-\eta} \times \frac{\eta^0}{0!} \times \mathbf{P} (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \\ &\quad + \mathbf{P} \times \exp(-\eta \mathbf{L}_m) \times (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \end{aligned} \quad (3)$$

where $\exp(-\eta \mathbf{L}_m) = e^{-\eta} \sum_{t=0}^{\infty} \left(\frac{\eta^t}{t!} (\mathbf{P}^T \mathbf{P})^t \right)$ and $\mathbf{L}_m = \mathbf{I}_m - \mathbf{P}^T \mathbf{P}$

Finally, we derive the heat kernel diffusion matrix in linear form [2]:

$$\mathbf{F} = e^{-\eta} \mathbf{I}_n - \hat{\mathbf{P}} \hat{\mathbf{P}}^T (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T, \quad (4)$$

where $\hat{\mathbf{P}} = e^{-\eta} \mathbf{I}_m - \exp(-\eta \mathbf{L}_m)$. $\square$

2 B: Proofs of Theorem 2

THEOREM 2. Let $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ with t -SVD $\mathcal{B} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$ and $\beta > 0$. For the Tensor Exponential Norm minimization TENM problem:

$$\arg \min_{\mathcal{A}} \frac{1}{2} \|\mathcal{A} - \mathcal{B}\|_F^2 + \beta \|\mathcal{A}\|_{\text{TEN}}. \quad (5)$$

The optimal solution $\mathcal{A}^*$ is given by

$$\mathcal{A}^* = \mathcal{U} * \text{ifft}(\text{Prox}_{f,\beta}(\mathcal{S}_f), [], 3) * \mathcal{V}^T, \quad (6)$$

where $\text{ifft}(\text{Prox}_{f,\beta}(\mathcal{S}_f), [], 3) \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f -diagonal tensor, with $P_\beta(\mathcal{S}_f^{(k)}(i, i))$ satisfying [4]:

$$\begin{aligned} \text{Prox}_{f,\beta}(\mathcal{S}_f^{(k)}(i, i)) &= \arg \min_{\sigma \geq 0} \frac{1}{2} (\sigma - \mathcal{S}_f^{(k)}(i, i))^2 \\ &\quad + \beta \left(1 - e^{-\frac{-\mathcal{S}_f^{(k)}(i, i)}{\delta}} \right), \end{aligned} \quad (7)$$

for $1 \leq k \leq n_3$ and $1 \leq i \leq \min(n_1, n_2)$. Notably, $\text{Prox}_{f,\beta}(\mathcal{S}_f^{(k)}(i, i)) \in [0, \mathcal{S}_f^{(k)}(i, i)]$, and $\text{Prox}_{f,\beta}(\mathcal{S}_f^{(k)}(i, i)) = 0$ if $\mathcal{S}_f^{(k)}(i, i) = 0$.

To prove Theorem 2, we first introduce an important lemma. Specially, we consider the Tensor Exponential Norm minimization (TENM) problem in the complex matrix case.

LEMMA 3. [3] Given $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$, and $\mathbf{B} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ is the SVD of $\mathbf{B}$ and $\beta > 0$, the complex matrix exponential norm minimization:

$$\arg \min_{\mathbf{A}} \frac{1}{2} \|\mathbf{A} - \mathbf{B}\|_F^2 + \beta \|\mathbf{A}\|_{\text{TEN}}. \quad (8)$$

has an optimal solution:

$$\mathbf{A}^* = \mathbf{U}_B \text{Prox}_{f,\beta}(\mathbf{S}_B) \mathbf{V}_B^H, \quad (9)$$

where $\text{Prox}_{f,\beta}(\mathbf{S}_B)$ is a diagonal matrix with elements:

$$\text{Prox}_{f,\beta}(\mathbf{S}_B(i, i)) = \arg \min_{\sigma \geq 0} \frac{1}{2} (\sigma - \mathbf{S}_B(i, i))^2 + \beta (1 - e^{-\frac{\sigma}{\delta}}), \quad (10)$$

for $1 \leq i \leq \min(m, n)$.

Now we proceed to prove Theorem 2

PROOF. In the Fourier domain, according to the linearity of the FFT and the fact that $\|\mathcal{X}\|_F^2 = \frac{1}{n_3} \|\mathcal{X}_f\|_F^2$, we can reformulate the objective function as:

$$\begin{aligned} & \frac{1}{2} \|\mathcal{A} - \mathcal{B}\|_F^2 + \beta \|\mathcal{A}\|_{\text{TEN}} \\ &= \frac{1}{2n_3} \|\mathcal{A}_f - \mathcal{B}_f\|_F^2 + \frac{\beta}{n_3} \sum_{k=1}^{n_3} \|\mathcal{A}_f^{(k)}\|_{\text{TEN}} \\ &= \frac{1}{n_3} \sum_{k=1}^{n_3} \left(\frac{1}{2} \|\mathcal{A}_f^{(k)} - \mathcal{B}_f^{(k)}\|_F^2 + \beta \|\mathcal{A}_f^{(k)}\|_{\text{TEN}} \right), \end{aligned} \quad (11)$$

for for $1 \leq k \leq n$.

This allows us to decouple the tensor optimization problem into n_3 independent complex matrix optimization problems, one for each slice:

$$\arg \min_{\mathcal{A}_f^{(k)}} \frac{1}{2} \|\mathcal{A}_f^{(k)} - \mathcal{B}_f^{(k)}\|_F^2 + \beta \|\mathcal{A}_f^{(k)}\|_{\text{TEN}}, \quad (12)$$

Given the SVD of $\mathcal{B}_f^{(k)}$ as $\mathcal{B}_f^{(k)} = \mathcal{U}_f^{(k)} \mathcal{S}_f^{(k)} (\mathcal{V}_f^{(k)})^H$, we can apply Lemma 3, to find the optimal solution of 12:

$$\mathcal{A}_f^{*(k)} = \mathcal{U}_f^{(k)} \text{Prox}_{f,\beta}(\mathcal{S}_f^{(k)}) (\mathcal{V}_f^{(k)})^H, \quad (13)$$

where $P_\beta(\mathcal{S}_f^{(k)}(i, i))$ is obtained by solving:

$$\arg \min_{\sigma \geq 0} \frac{1}{2} (\sigma - \mathcal{S}_f^{(k)}(i, i))^2 + \beta(1 - e^{-\frac{\sigma}{\delta}}). \quad (14)$$

Note that both $1 - e^{-\frac{\sigma}{\delta}}$ and $\frac{1}{2}(\sigma - \mathcal{S}_f^{(k)}(i, i))^2$ are non-decreasing for $\sigma > \mathcal{S}_f^{(k)}(i, i)$. Therefore, $\text{Prox}_{f,\beta}(\mathcal{S}_f^{(k)}(i, i)) \in [0, \mathcal{S}_f^{(k)}(i, i)]$. Additionally, if $\mathcal{S}_f^{(k)}(i, i) = 0$, then $P_\beta(\mathcal{S}_f^{(k)}) = 0$.

Finally, to obtain the optimal solution $\mathcal{A}_f^*$ by 12, we apply the inverse FFT operator:

$$\mathcal{A}^* = \text{ifft}(\mathcal{A}_f^*, [], 3) = \mathcal{U} \times \text{ifft}(\text{Prox}_{f,\beta}(\mathcal{S}_f), [], 3) \times \mathcal{V}^T \quad (15)$$

□

3 C: Proofs of Theorem 3

THEOREM 4. Let $\{\mathcal{P}_k = (\mathbf{H}_k^r, \mathbf{E}_k^r, \mathcal{Y}_k, \mathbf{W}_k^r, \mathcal{G}_k)\}_{k=1}^\infty$ be the sequence generated by the Algorithm HBCD, then the sequence $\{\mathcal{P}_k\}$ meets the following two principles:

- $\{\mathcal{P}_k\}$ is bounded;
- Any accumulation point of $\{\mathcal{P}_k\}$ is a KKT point of the Algorithm HBCD.

To prove Theorem 4, we first introduce two important lemmas.

LEMMA 5. [6] Let $\mathcal{H}$ be a real Hilbert space endowed with an inner product $\langle \cdot, \cdot \rangle$, a norm $\|\cdot\|$ with the dual norm $\|\cdot\|^{dual}$ and $y \in \partial\|x\|$, where $\partial f(\cdot)$ is the subgradient of $f(\cdot)$. Then $\|y\|^{dual} = 1$ if $x \neq 0$ and $\|y\|^{dual} \leq 1$ if $x = 0$.

LEMMA 6. [5] Suppose that $\mathbf{F} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is defined as $\mathbf{F}(\mathbf{X}) = f \circ \sigma(\mathbf{X}) = f(\sigma_1(\mathbf{X}), \dots, \sigma_h(\mathbf{X}))$, where $\mathbf{X} = \mathbf{U} \text{Diag}(\sigma(\mathbf{X})) \mathbf{V}^T$ is the SVD of matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$, $h = \min(m, n)$ and $f(\cdot) : \mathbb{R}^h \rightarrow \mathbb{R}$ is differentiable and absolutely symmetric at $\sigma(\mathbf{X})$. Then, the subdifferential

of $F(\mathbf{X})$ at $\mathbf{X}$ is

$$\frac{\partial F(\mathbf{X})}{\partial \mathbf{X}} = \mathbf{U} \text{Diag}(\partial f(\sigma(\mathbf{X}))) \mathbf{V}^T, \quad (16)$$

where $\partial f(\sigma(\mathbf{X})) = (\frac{\partial f(\sigma_1(\mathbf{x}))}{\partial \mathbf{x}}, \dots, \frac{\partial f(\sigma_h(\mathbf{x}))}{\partial \mathbf{x}})$.

PROOF. Proof of 1st part: On the $k+1$ iteration, from the updating rule of $\mathbf{E}_{k+1}^r$, the first-order optimal condition should be satisfied.

$$\begin{aligned} 0 &= \lambda \partial \|\mathbf{E}_{k+1}^r\|_{2,1} \\ &+ \mu_k (\mathbf{E}_{k+1}^r - (\widetilde{\mathbf{B}}^r - \mathbf{H}_{k+1}^r + \mathbf{S}_k^r / \mu_k)) \\ &= \lambda \partial \|\mathbf{E}_{k+1}^r\|_{2,1} - \mathbf{S}_{k+1}^r, \end{aligned} \quad (17)$$

where $\widetilde{\mathbf{B}}^r = \frac{1}{r-1} \sum_{k \neq r}^v (\mathbf{F}^k \mathbf{B}^r)$.

Thus, we have

$$\frac{1}{\lambda} [\mathbf{S}_{k+1}^r]_{:,j} = \partial \|\mathbf{E}_{k+1}^r\|_{2,1}, \quad (18)$$

where $[\mathbf{S}_{k+1}^r]_{:,j}$ and $[\mathbf{E}_{k+1}^r]_{:,j}$ are the j -th columns of $\mathbf{S}_{k+1}^r$ and $\mathbf{E}_{k+1}^r$, respectively. Since the ℓ_2 norm is self-dual, by Lemma 6, we have $\|\frac{1}{\lambda} [\mathbf{S}_{k+1}^r]_{:,j}\|_2 \geq 1$. Therefore, the sequence $\{\mathbf{S}_{k+1}^r\}$ is bounded. Next, we prove that the sequence $\{\mathcal{Y}_{k+1}\}$ is bounded. According to the updating rule of $\mathcal{G}$, the first-order optimality condition gives

$$\partial \|\mathcal{G}_{k+1}\|_{\text{TEN}} = \mathcal{Y}_{k+1}. \quad (19)$$

Let $\mathbf{U} * \mathbf{S} * \mathbf{V}^T$ be the t -SVD of tensor $\mathcal{G}$. By Lemma 6 and definition of TEN, we have:

$$\begin{aligned} & \|\partial \|\mathcal{G}_{k+1}\|_{\text{TEN}}\|_F^2 \\ &= \left\| \frac{1}{n_3} \mathcal{U} * \text{ifft}(\partial(\mathcal{S}_f), [], 3) * \mathcal{V}^T \right\|_F^2 \\ &= \frac{1}{n_3^2} \|\partial f(\mathcal{S}_f)\|_F^2 \\ &\leq \frac{1}{n_3^2} \sum_{i=1}^{n_3} \sum_{j=1}^{\min(n_1, n_2)} [\partial f(\mathcal{S}_f^{(i,j)})]^2 \\ &\leq \frac{e^{-\frac{2}{\delta}} \min(n_1, n_2)}{\delta^2 n_3^2} \end{aligned} \quad (20)$$

where the second inequality follows from $\partial f(x) \leq \frac{e^{-\frac{1}{\delta}}}{\delta}$, and $f(x) = 1 - e^{-\frac{x}{\delta}}$ is our improved tensor. Thus $\partial \|\mathcal{G}_{k+1}\|_{\text{TEN}}$ is bounded, and consequently, the sequence $\{\mathcal{S}_{k+1}\}$ is also bounded. From the iterative method in the algorithm for solving HBCD, we can deduce:

$$\begin{aligned} & \mathcal{L}_{\mu_k, \rho_k}(\mathbf{H}_{k+1}^r, \mathbf{E}_{k+1}^r, \mathcal{G}_{k+1}, \mathbf{S}_k^r, \mathcal{Y}_k) \\ &\leq \mathcal{L}_{\mu_k, \rho_k}(\mathbf{H}_k^r, \mathbf{E}_k^r, \mathcal{G}_k, \mathbf{S}_k^r, \mathcal{Y}_k) \\ &= \mathcal{L}_{\mu_{k-1}, \rho_{k-1}}(\mathbf{H}_k^r, \mathbf{E}_k^r, \mathcal{G}_k, \mathbf{S}_{k-1}^r, \mathcal{Y}_{k-1}) \\ &+ \frac{\rho_k + \rho_{k-1}}{2\rho_{k-1}^2} \|\mathcal{Y}_k - \mathcal{Y}_{k-1}\|_F^2 \\ &+ \frac{\mu_k + \mu_{k-1}}{2\mu_{k-1}^2} \sum_{v=1}^m \|\mathbf{S}_k^r - \mathbf{S}_{k-1}^r\|_F^2, \end{aligned} \quad (21)$$

Summing both sides of the inequality 21 form $k = 1$ to n , we get:

$$\begin{aligned} & \mathcal{L}_{\mu_k, \rho_k}(\mathbf{H}_{k+1}^r, \mathbf{E}_{k+1}^r, \mathcal{G}_{k+1}, \mathbf{S}_k^r, \mathcal{Y}_k) \\ & \leq \mathcal{L}_{\mu_0, \rho_0}(\mathbf{H}_1^r, \mathbf{E}_1^r, \mathcal{G}_1, \mathbf{S}_0^r, \mathcal{Y}_0) \\ & + \sum_{k=1}^n \left(\frac{\rho_k + \rho_{k-1}}{2\rho_{k-1}^2} \|\mathcal{Y}_k - \mathcal{Y}_{k-1}\|_F^2 \right) \\ & + \sum_{k=1}^n \left(\frac{\mu_k + \mu_{k-1}}{2\mu_{k-1}^2} \sum_{v=1}^m \|\mathbf{S}_k^r - \mathbf{S}_{k-1}^r\|_F^2 \right) \end{aligned} \quad (22)$$

Observe that

$$\sum_{k=1}^n \frac{\mu_k + \mu_{k+1}}{2\mu_{k-1}^2} < \infty, \quad \sum_{k=1}^n \frac{\rho_k + \rho_{k+1}}{2\rho_{k-1}^2} < \infty \quad (23)$$

Note that $\mathcal{L}_{\mu_0, \rho_0}(\mathbf{H}_1^r, \mathbf{E}_1^r, \mathcal{G}_1, \mathbf{S}_0^r, \mathcal{Y}_0)$ is finite, and the sequence $\{\mathbf{Y}_k^r\}$, $\mathcal{W}_k$, $\sum_{k=1}^n \frac{\mu_k + \mu_{k+1}}{2\mu_{k-1}^2}$ and $\sum_{k=1}^n \frac{\rho_k + \rho_{k+1}}{2\rho_{k-1}^2}$ are all bounded. Therefore $\mathcal{L}_{\mu_k}(\mathbf{H}_{k+1}^r, \mathbf{E}_{k+1}^r, \mathcal{G}_{k+1}, \mathbf{S}_k^r, \mathcal{Y}_k)$ is bounded. Expanding $\mathcal{L}_{\mu_k, \rho_k}$, we have:

$$\begin{aligned} & \mathcal{L}_{\mu_k, \rho_k}(\mathbf{H}_{k+1}^r, \mathbf{E}_{k+1}^r, \mathcal{G}_{k+1}, \mathbf{S}_k^r, \mathcal{Y}_k) \\ & = \|\mathcal{G}_{k+1}\|_{\text{TEN}} + \lambda \|\mathbf{E}_{k+1}\|_{2,1} \\ & + \sum_{r=1}^v \langle \mathbf{S}_k^r, \tilde{\mathbf{B}}^r - \mathbf{H}_{k+1}^r - \mathbf{E}_{k+1}^r \rangle \\ & + \frac{\mu_k}{2} \|\tilde{\mathbf{B}}^r - \mathbf{H}_{k+1}^r - \mathbf{E}_{k+1}^r\|_F^2 \\ & + \langle \mathcal{Y}_k, \mathcal{H}_{k+1} - \mathcal{G}_{k+1} \rangle + \frac{\rho_k}{2} \|\mathcal{H}_{k+1} - \mathcal{G}_{k+1}\|_F^2. \end{aligned} \quad (24)$$

Since each term in Eq. 24 is non-negative and $\mathcal{L}_{\mu_k}(\mathbf{H}_{k+1}^r, \mathbf{E}_{k+1}^r, \mathcal{G}_{k+1}, \mathbf{S}_k^r, \mathcal{Y}_k)$ is bounded, we can deduce that each term in Eq. 24 is bounded. The boundedness of $\|\mathcal{G}_{k+1}\|_{\text{TEN}}$ implies that all singular values of $\mathcal{G}_k + 1$ are bounded. Furthermore, based on the following equation

$$\|\mathcal{G}_{k+1}\|_F^2 = \frac{1}{n_3} \|\mathcal{G}_{f,k+1}\|_F^2 = \frac{1}{n_3} \sum_{i=1}^{n_3} \sum_{j=1}^{\min(n_1, n_2)} (\mathbf{S}_f^i(j, j))^2, \quad (25)$$

we can derive the sequence $\mathcal{G}_{k+1}$ is bounded, then, it is easy to prove the boundedness of $\mathbf{H}_{k+1}^r$. Therefore, we can conclude that the sequence $\{\mathcal{P}_k = (\mathbf{H}_k^r, \mathbf{E}_k^r, \mathcal{Y}_k, \mathbf{W}_k^r, \mathcal{G}_k)\}_{k=1}^\infty$ generated by the Algorithm HBCD is bounded. $\square$

PROOF. Proof of 2nd part: By the Weierstrass Bolzano theorem [1], there exists at least one accumulation point of the sequence $\{\mathcal{P}_k = (\mathbf{H}_k^r, \mathbf{E}_k^r, \mathcal{Y}_k, \mathbf{W}_k^r, \mathcal{G}_k)\}_{k=1}^\infty$. Let us denote one such point as $\mathcal{P}_*$. Then we have

$$\lim_{k \rightarrow \infty} (\mathbf{H}_k^r, \mathbf{E}_k^r, \mathbf{S}_k^r, \mathcal{Y}_k, \mathcal{G}_k) = (\mathbf{H}_*^r, \mathbf{E}_*^r, \mathbf{S}_*^r, \mathcal{Y}_*, \mathcal{G}_*). \quad (26)$$

Form the updating rule of $\mathbf{S}_k^r$ and $\mathcal{Y}_k$, we obtain the following equations:

$$\begin{aligned} \tilde{\mathbf{B}}^r - \mathbf{H}_{k+1}^r - \mathbf{E}_{k+1}^r &= (\mathbf{S}_{k+1}^r - \mathbf{S}_k^r) / \mu_t, \\ \mathcal{H}_{k+1} - \mathcal{G}_{k+1} &= (\mathcal{Y}_{k+1} - \mathcal{Y}_k) / \rho_t. \end{aligned} \quad (27)$$

Given the boundedness of sequences $\mathbf{S}_k^r$ and $\mathcal{Y}_k$, and the fact $\lim_{k \rightarrow \infty} \mu_k = \infty$, we have:

$$\begin{aligned} \lim_{k \rightarrow \infty} \tilde{\mathbf{B}}^r - \mathbf{H}_{k+1}^r - \mathbf{E}_{k+1}^r &= \lim_{k \rightarrow \infty} (\mathbf{S}_{k+1}^r - \mathbf{S}_k^r) / \mu_t = 0, \\ \lim_{k \rightarrow \infty} \mathcal{H}_{k+1} - \mathcal{G}_{k+1} &= \lim_{k \rightarrow \infty} (\mathcal{Y}_{k+1} - \mathcal{Y}_k) / \rho_t = 0, \end{aligned} \quad (28)$$

Consequently, we can conclude:

$$\tilde{\mathbf{B}}^r - \mathbf{H}_*^r - \mathbf{E}_*^r, \quad \mathcal{H}_* - \mathcal{G}_* = 0. \quad (29)$$

Furthermore, due to the first-order optimality conditions of $\mathbf{E}_{k+1}^r$ and $\mathbf{G}_{k+1}^r$, we can deduce

$$\begin{aligned} 0 &= \alpha \partial \|\mathbf{E}_{k+1}^r\|_{2,1} - \mathbf{S}_{k+1}^r \Rightarrow \mathbf{S}_*^r = \alpha \partial \|\mathbf{E}_*^r\|_{2,1} \\ 0 &= \partial \|\mathcal{G}_{k+1}\|_{\text{TEN}} - \mathcal{Y}_{k+1} \Rightarrow \mathcal{Y}_* = \partial \|\mathcal{G}_*\|_{\text{TEN}} \end{aligned} \quad (30)$$

Thus, the accumulation point $\mathcal{P}_*$ of the sequence $\{\mathcal{P}_k = (\mathbf{H}_k^r, \mathbf{E}_k^r, \mathcal{Y}_k, \mathbf{W}_k^r, \mathcal{G}_k)\}_{k=1}^\infty$ generated by the algorithm of solving HBCD satisfies the KKT condition. $\square$

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